

Duy Tran

337-517-5684 | duyatran05@gmail.com | [linkedin.com/in/duy-a-tran](https://www.linkedin.com/in/duy-a-tran) | github.com/itssduy | duyatran.com

Education

University of Louisiana at Lafayette

B.S. in Computer Science, Cloud Computing Concentration, GPA: 3.5/4.0

Dec 2026

Certifications

- CompTIA Security+ Expected Feb 2027
- AWS Certified Developer - Associate Expected Dec 2026
- AWS Certified Solutions Architect - Associate Sept 2026
- AWS Certified Cloud Practitioner Mar 2026
- Harvard CS50 Jan 2025

Experience

CGI

Jun 2026 - Present

Cloud Developer Intern

Lafayette, LA

- Enhanced an internal AWS auto-tagging system, **deployed across multiple orgs and partitions**, using EventBridge, Lambda, and cross-account IAM to track resource ownership for Bedrock AgentCore.
- Migrated a multi-cloud AI pricing assistant's frontend from Streamlit to Chainlit **one month ahead of schedule**, integrating an orchestrator with Strands for sub-agents executing MCP tools to **add real-time streaming for beta release**.
- Architected a 9+ module Terraform library for Bedrock AgentCore adopted as the standard across a 40+ person cloud engineering team for internal and MSP efforts.
- Used AI-assisted, spec-driven development in devcontainers with GitLab CI/CD **cutting average feature delivery time by ~30%**.

FIPSE

April 2026 - Present

Software Engineer

Lafayette, LA

- Built a web application enabling **400+ educators and students** to create and evaluate AI learning agents through a Gem builder.
- Containerized the platform (Django, PostgreSQL, Caddy) with Docker Compose for consistent local/prod environments, managing sprints and PR reviews via Azure DevOps.
- Built a Gemini-powered agent application that answers both natural-language and analytical queries over **200+ faculty CVs**, using RAG and structured JSON extraction to make grant data searchable.
- Structured output into searchable JSON, cutting document review time from **hours to minutes per file**.
- Configured GCP IAM roles and service account credentials, and stored extracted data in Cloud Storage buckets.

CAPE

Aug 2024 - April 2026

Lead Frontend Engineer & Full-Stack Developer

Lafayette, LA

- Architected ESG-Grid, a patented, ITAR-compliant, multi-tenant ground-station web interface in Angular, modeling telemetry across organizations, missions, vehicles, and payload subsystems, **backed by the Governor of Louisiana**.
- Built full-stack APIs with Django Ninja and PostgreSQL, API key auth, and containerized backend workloads on Oracle Government Cloud.
- **Led a team of 5 frontend developers within a 100+ member org**, managing sprints, PR reviews, and compliance-driven code in Bitbucket.
- Implemented frontend development standards that **reduced bugs by ~20%** and improved development velocity across the team.
- **Awarded 2nd Place** at the Undergraduate Research Conference, University of Louisiana at Lafayette.

Projects

Cloud Cost Optimization Platform

Aug 2026 - Present

- Built a serverless AWS cost-optimization platform using EventBridge and Lambda to pull cost/utilization data from Cost Explorer and CloudWatch, storing results in DynamoDB.
- Built an EWMA-based rule engine to automatically detect and alert on idle AWS resources (EC2, EBS, snapshots) via SNS, **flagging findings across 100+ monitored resources**.
- Deployed a dashboard (ECS Fargate, API Gateway, Lambda) via Terraform, **automating cleanup for an estimated 80% of flagged findings**.

Personal Learning Agent

June 2026 - Jul 2026

- Designed a RAG-powered learning agent (Strands Agents SDK + Claude) with guided prompt logic that tests understanding rather than returning flat answers, using similarity thresholds to reduce hallucinations.
- Built an ingestion pipeline extracting text from files, URLs, and S3 URIs into S3 Vectors with Titan embeddings and source citations, plus a weekly refresh pipeline via EventBridge/Lambda/Terraform.
- Processed and indexed **100+ pages** of source material into S3 Vectors

Technical Skills

Languages: Python, C, Java, JavaScript, TypeScript, HTML, CSS

Frameworks: React, Angular, Express, Django, Flask, FastAPI, Chainlit

Databases: PostgreSQL, SQLite, MongoDB, DynamoDB, RDS

Cloud: AWS (Bedrock, EC2, Lambda, EventBridge, S3, ECR, ECS, DynamoDB, IAM, VPC, CloudTrail, SNS), Oracle, GCP

Tools: Git, Docker, Jira, Linux, Postman, CI/CD, Figma, Terraform, Azure DevOps, ServiceNow, Strands